Exercise 4

Define an RNN for regression, having 2 inputs, a hidden state of dimension 3, and 2 outputs. The activation function for the hidden state is ReLU. Write the equations for two time steps of forward pass through the network. For the inputs $\boldsymbol{X}_1 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$, $\boldsymbol{X}_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, if the weight matrices and bias vector are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the outputs $\boldsymbol{O}_1$ and $\boldsymbol{O}_2$ of the network. Specify the total number of real parameters (weights and biases) for the layer.

Solution:

we have the following formulas:

$$\boldsymbol{H}_t = \phi(\boldsymbol{X}_t \boldsymbol{W}_{xh} + \boldsymbol{H}_{t-1} \boldsymbol{W}_{hh} + \boldsymbol{b}_h),$$

$$\boldsymbol{O}_t = \boldsymbol{H}_t \boldsymbol{W}_{hq} + \boldsymbol{b}_q,$$

where $\boldsymbol{X}_t \in \mathbb{R}^{n \times d}$, $\boldsymbol{W}_{xh} \in \mathbb{R}^{d \times h}$, $\boldsymbol{W}_{hh} \in \mathbb{R}^{h \times h}$, $\boldsymbol{b}_h \in \mathbb{R}^{1 \times h}$, $\boldsymbol{H}_t \in \mathbb{R}^{n \times h}$, $\boldsymbol{W}_{hq} \in \mathbb{R}^{h \times q}$, $\boldsymbol{b}_q \in \mathbb{R}^{1 \times q}$, $\boldsymbol{O}_t \in \mathbb{R}^{n \times q}$, and

$$\phi(x) = \text{ReLU}(x)$$

in our case, $n = 2$, $d = 2$, $h = 3$, $q = 2$

we take

$$\boldsymbol{W}_{xh} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3},$$

$$\boldsymbol{W}_{hh} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 3},$$

$$\boldsymbol{b}_h = \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} \in \mathbb{R}^{1 \times 3},$$

$$\boldsymbol{W}_{hq} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2},$$

$$\boldsymbol{b}_q = \begin{bmatrix} 1 & -1 \end{bmatrix} \in \mathbb{R}^{1 \times 2},$$

$$\boldsymbol{H}_0 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$$

- now, we apply the formulas:

$$\begin{aligned} \boldsymbol{H}_1 &= \phi\left(\begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 2 & 0 & 0 \\ 2 & -1 & 0 \end{bmatrix}\right) \\ &= \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3}, \end{aligned}$$

$$\begin{aligned} \boldsymbol{H}_2 &= \phi\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 & -2 \\ 2 & 0 & -2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}\right) \\ &= \phi\left(\begin{bmatrix} 3 & 1 & -3 \\ 4 & 0 & -2 \end{bmatrix}\right) \\ &= \begin{bmatrix} 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3}, \end{aligned}$$

H_1 W_{hq} b_q

$t=1$

$$
\begin{aligned}
o_1 &= \begin{bmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

H_2 W_{hq} b_q

$t=2$

$$
\begin{aligned}
o_2 &= \begin{bmatrix} 3 & 1 & 0 \\ 4 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 1 \\ 4 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & 0 \\ 5 & -1 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

$$
\begin{aligned}
\text{Total parameters} &= 2\times 3+3\times 3+1\times 3+3\times 2+1\times 2 \\
&= 6+9+3+6+2 \\
&= 26.
\end{aligned}
$$

W_{xh} W_{hh} b_h W_{hq} b_q

$$\text{param} = d\times h + h\times h + 1\times h + h\times q + 1\times q = 26$$

(2.) Ex de la curs S11

GRU

$$x_1 = \begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 2} \qquad x_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in \mathbb{R}^{2\times 2}$$

$h = 3$, $d = 2$ (coloane), $n = 2$ (nr de exe) (linii), $T = 2$

$$W_{xh} = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{d\times h}_{2\times 3}$$

$$W_{xz} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{d\times h}_{2\times 3}$$

$$W_{hr} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{h\times h}_{3\times 3}$$

$$W_{hz} = \begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \in \mathbb{R}^{h\times h}_{3\times 3}$$

$$b_h = \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} \in \mathbb{R}^{1\times h}_{1\times 3}$$

$$b_z = \begin{pmatrix} 0 & -1 & 1 \end{pmatrix} \in \mathbb{R}^{1\times h}_{1\times 3}$$

$$W_{xh} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{d\times h}_{2\times 3}$$

$$W_{hh} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \in \mathbb{R}^{h\times h}_{3\times 3}$$

$b_h = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{1\times 3}$

$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in \mathbb{R}^{2\times 3}$

- $t = 1$

$R_1 = \sigma(x_1 W_{xh} + H_0 W_{hh} + b_h) =$

$= \sigma\left(\begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix} + \right.$

$\left. + \begin{pmatrix} 1 & 0 & -1 \end{pmatrix}\right) =$

$= \sigma\left(\begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \end{pmatrix} + 0^{2\times 3} + \begin{pmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}\right) =$

$= \begin{pmatrix} \sigma(2) & \sigma(-1) & \sigma(0) \\ \sigma(2) & \sigma(0) & \sigma(-1) \end{pmatrix}$

$z_1 = \sigma(x_1 W_{xz} + H_0 W_{hz} + b_z) =$

$= \sigma\left(\begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix} + 0^{2\times 3}\begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -1 & 1 \end{pmatrix}\right) =$

după calcule $= \begin{pmatrix} \sigma(-1) & \sigma(-1) & \sigma(3) \\ \sigma(0) & \sigma(1) & \sigma(2) \end{pmatrix}$

$$\tilde{H}_1 = \tanh\left(x_1 W_{xh} + (R_1 \odot H_0) W_{hh} + b_h\right) =$$

$$= \tanh\left(\begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \end{pmatrix} + \left(\begin{pmatrix} \nabla(2) & \nabla(-1) & \nabla(0) \\ \nabla(2) & \nabla(0) & \nabla(-1) \end{pmatrix} \odot 0^{2\times 3}\right)\right.$$

$$\left.\begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}\right) =$$

$$= \tanh\left(\begin{pmatrix} 2 & 1 & -1 \\ 2 & 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} \tanh(2) & \tanh(1) & \tanh(-1) \\ \tanh(2) & \tanh(0) & \tanh(0) \end{pmatrix}$$

nr de param = $d\times h + d\times h + h\times h + h\times h + 1\times h + 1\times h + d\times h + h\times h + 1\times h = 6+6+9+9+3+3+6+9+3 = 54$

$$H_1 = z_1 \odot H_0 + (1 - z_1) \odot \tilde{H}_1 =$$

$$= \begin{pmatrix} \nabla(-1) & \nabla(-1) & \nabla(3) \\ \nabla(0) & \nabla(-1) & \nabla(2) \end{pmatrix} \odot 0^{2\times 3} + \begin{pmatrix} 1-\nabla(-1) & 1-\nabla(-1) & 1-\nabla(3) \\ 1-\nabla(0) & 1-\nabla(-1) & 1-\nabla(2) \end{pmatrix} \odot$$

$$\odot \begin{pmatrix} \tanh(2) & \tanh(1) & \tanh(-1) \\ \tanh(2) & \tanh(0) & \tanh(0) \end{pmatrix} =$$

$$0^{2\times 3} + [\ldots]$$

După se fac aceeași pași și pt $t = 2$

3.

Question 8
Not yet answered
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Flag question

Compute symbolically the total number of parameters of a GRU layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

Maximum file size: 5MB, maximum number of files: 1

$n = 2 \quad d = 3 \quad h = 4$

$X_t \in \mathbb{R}^{n \times d} = \mathbb{R}^{2 \times 3}$

$H_{t-1} \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4}$

$R_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \quad R_t = \sigma(X_t W_{xh} + H_{t-1} W_{hh} + b_h)$

$Z_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \quad Z_t = \sigma(X_t W_{xz} + H_{t-1} W_{hz} + b_z)$

Parametrii

$W_{xh}, W_{xz} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \Rightarrow 2 \cdot (3 \cdot 4) = 24$

$W_{hh}, W_{hz} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \Rightarrow 2 \cdot (4 \cdot 4) = 32$

$b_h, b_z \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \Rightarrow 2 \cdot (1 \cdot 4) = 8$

$\tilde{H}_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4}$

64

$\tilde{H}_t = \tanh(X_t W_{xh} + (R_t \odot H_{t-1}) W_{hh} + b_h)$

Parametrii:

$W_{xh} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \Rightarrow 3 \cdot 4 = 12$

$W_{hh} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \Rightarrow 4 \cdot 4 = 16$

$b_h \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \Rightarrow 1 \cdot 4 = 4$

32

$64 + 32 = 96$

4. Compute symbolically the total number of parameters of an LSTM layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: n=2, d=3, h=4.

Compute symbolically the total number of parameters of an LSTM layer, knowing that the number of examples is n, the number of inputs is d, and the number of hidden units is h. Numerical application: $n = 2, d = 3, h = 4$.

$n = 2 \quad d = 3 \quad h = 4$

$x_t \in \mathbb{R}^{n \times d} = \mathbb{R}^{2 \times 3}$

$H_{t-1} \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4}$

$I_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \qquad I_t = \sigma(x_t W_{xi} + H_{t-1} W_{hi} + b_i)$

$F_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \qquad F_t = \sigma(x_t W_{xf} + H_{t-1} W_{hf} + b_f)$

$O_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4} \qquad O_t = \sigma(x_t W_{xo} + H_{t-1} W_{ho} + b_o)$

Param:

$W_{xi}, W_{xf}, W_{xo} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \Rightarrow 3 \cdot (4 \cdot 3) = 36$

$W_{hi}, W_{hf}, W_{ho} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \Rightarrow 3 \cdot (4 \cdot 4) = 48$

$b_i, b_f, b_o \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \Rightarrow 3 (1 \cdot 4) = 12$

96

$\tilde{C}_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{2 \times 4}$

$\tilde{C}_t = \tanh(x_t W_{xc} + H_{t-1} W_{hc} + b_c)$

$W_{xc} \in \mathbb{R}^{d \times h} = \mathbb{R}^{3 \times 4} \Rightarrow 3 \cdot 4 = 12$

$W_{hc} \in \mathbb{R}^{h \times h} = \mathbb{R}^{4 \times 4} \Rightarrow 4 \cdot 4 = 16$

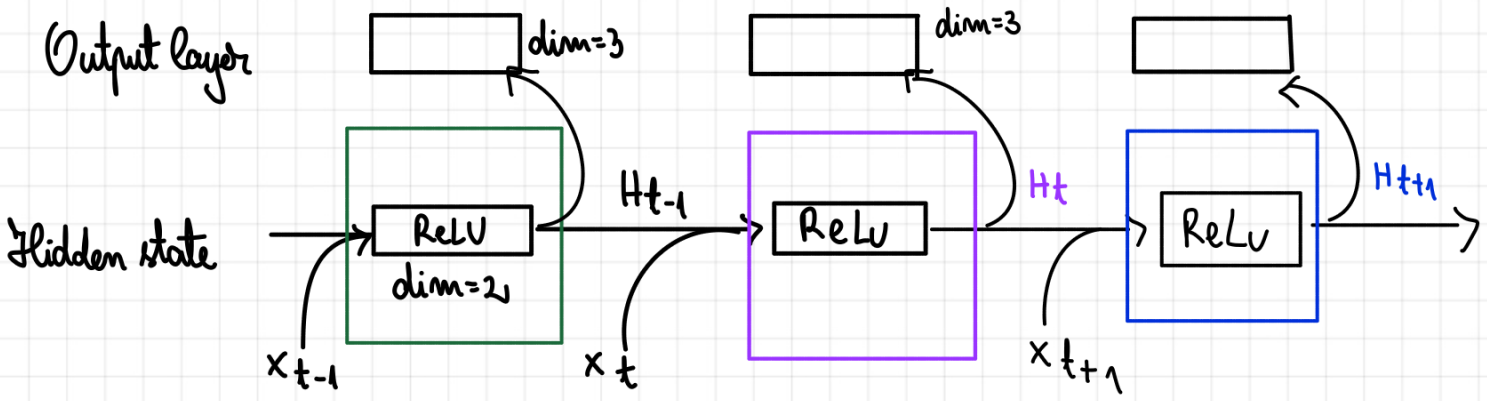
$b_c \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 4} \Rightarrow 1 \cdot 4 = 4$

32

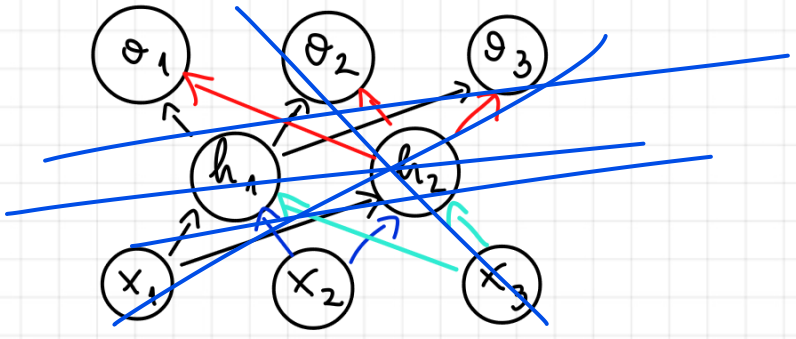
param = 96 + 32 = 128

5.

4. Draw the diagram of an RNN for regression, having 3 inputs, a hidden state of dimension 2, and 3 outputs. The activation function for the hidden state is ReLU. For the inputs $\boldsymbol{X}_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $\boldsymbol{X}_2 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$, if the weight matrices and bias vector are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the outputs $\boldsymbol{O}_1$ and $\boldsymbol{O}_2$ of the network. Specify the total number of real parameters (weights and biases) for the network.



pt un mom de timp



$n = 2$, $d = 3$ (nr inputuri), $h = 2$, $q = 3$

Solution:
we have the following formulas:

$$\boldsymbol{H}_t = \phi(\boldsymbol{X}_t \boldsymbol{W}_{xh} + \boldsymbol{H}_{t-1} \boldsymbol{W}_{hh} + \boldsymbol{b}_h),$$

$$\boldsymbol{O}_t = \boldsymbol{H}_t \boldsymbol{W}_{hq} + \boldsymbol{b}_q,$$

where $\boldsymbol{X}_t \in \mathbb{R}^{n \times d}$, $\boldsymbol{W}_{xh} \in \mathbb{R}^{d \times h}$, $\boldsymbol{W}_{hh} \in \mathbb{R}^{h \times h}$, $\boldsymbol{b}_h \in \mathbb{R}^{1 \times h}$, $\boldsymbol{H}_t \in \mathbb{R}^{n \times h}$, $\boldsymbol{W}_{hq} \in \mathbb{R}^{h \times q}$, $\boldsymbol{b}_q \in \mathbb{R}^{1 \times q}$, $\boldsymbol{O}_t \in \mathbb{R}^{n \times q}$, and

$$\phi(x) = \text{ReLU}(x)$$

W_{xh} (3×2) $= \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$

W_{hh} (2×2) $= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ W_{hq} (2×3) $= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b_h (1×2) $= (1 \quad 0)$

b_q (1×3) $= (1 \quad 0 \quad 0)$

$$H_0 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$
2×2

$$H_1 = \tilde{V}\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} + 0^{2\times 2} W_{hh} + \begin{pmatrix} 1 & 0 \end{pmatrix} \right) =$$

$$= \tilde{V}\left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$$
relu

$$O_1 = H_1 W_{hq} + b_q =$$

$$= \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 3 & 0 & 0 \end{pmatrix}$$

-> la fel si pt H_2, O_2

$$\text{param} = 3\times 2 + 2\times 2 + 1\times 2 + 2\times 3 + 1\times 3 =$$

$$= 6 + 4 + 2 + 6 + 3 = 21$$

6.

- FC-N stands for a fully-connected layer with N units.

8. Write the equations for the forward pass of an RNN with two hidden states. Compute symbolically the total number of parameters (weights and biases) of an RNN with two hidden states, knowing that the number of examples is n, the number of inputs is d, the number of hidden units of both hidden states is h, and the number of outputs is q. Numerical application: $n = 2, d = 3, h = 4, q = 2$. For the numerical application, write one step through the RNN, with the input, weights, and biases generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros).

$l = 2$ (nr. of hidden layers)

$n = 2$, $d = 3$, $h = 4$, $q = 2$ $\quad X_i \in \mathbb{R}^{2\times 3}$

$X_t \in \mathbb{R}^{n\times d} = \mathbb{R}^{2\times 3}$

$H_t^{(l)} \in \mathbb{R}^{n\times h} = \mathbb{R}^{2\times 4}$

$O_t \in \mathbb{R}^{n\times q} = \mathbb{R}^{2\times 2}$

$H_t^{(0)} = X_t \Rightarrow H_t^{(l)} = \phi\left(H_t^{(l-1)} W_{xh}^{(l)} + H_{t-1}^{(l)} W_{hh}^{(l)} + b_h^{(l)}\right)$

$W_{xh}^{(l)} \in \mathbb{R}^{d\times h} = \mathbb{R}^{3\times 4} \longrightarrow 2\,(3\cdot 4) = 2\cdot 12 = 24$

$W_{hh}^{(l)} \in \mathbb{R}^{h\times h} = \mathbb{R}^{4\times 4} \longrightarrow 2\,(4\cdot 4) = 2\cdot 16 = 32$

$b_h^{(l)} \in \mathbb{R}^{1\times h} = \mathbb{R}^{1\times 4} \longrightarrow 2\cdot(1\cdot 4) = 2\cdot 4 = 8$

64

$O_t = H_t^{(L)} W_{hq} + b_q$

$W_{hq} \in \mathbb{R}^{h\times q} = \mathbb{R}^{4\times 2} \longrightarrow 4\cdot 2 = 8$

$b_q \in \mathbb{R}^{1\times q} = \mathbb{R}^{1\times 2} \longrightarrow 1\cdot 2 = 2$

10

$\Rightarrow$ param $= 64 + 10 = 74$

$x_i \in \mathbb{R}^{2\times 3}$ ⟶ luăm unul singur (un pas)

$$x_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$W_{hh}^{(1)} = I_4$ $\quad b_h^{(1)} = (1\ 0\ 0\ 0)$

$W_{hh}^{(2)} = I_4$ $\quad b_h^{(2)} = (-1\ 0\ 0\ 0)$

$$W_{xh}^{(1)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad W_{xh}^{(2)} = I_4$$

$$W_{hq} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \qquad b_q = (1\ \ 0)$$

ReLu

$H_t^{(0)} = x_1$ $\quad H_0^{(1)} = 0^{2\times 4}$ $\quad H_0^{(2)} = 0^{2\times 4}$

$$H_1^{(1)} = \text{ReLu}\left(x_1 \cdot W_{xh}^{(1)} + H_0^{(1)} W_{hh}^{(1)} + b_h^{(1)}\right) =$$

$$= \text{ReLu}\left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + 0^{2\times 4} + (1\ 0\ 0\ 0)\right) =$$

$$= \text{ReLu}\left(\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}\right) =$$

$$= \begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 4}$$

$$H_1^{(2)} = \mathrm{ReLu}\left(H_1^{(1)} W_{xh}^{(2)} + H_0^{(2)} W_{hh}^{(2)} + b_h^{(2)} \right) =$$

$$= \mathrm{ReLu}\left(\begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} y_4 + 0_{2\times 4} \cdot y_4 + \begin{pmatrix} -1 & 0 & 0 & 0 \end{pmatrix} \right) =$$

$$= \mathrm{ReLu}\left(\begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \right) =$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{2\times 4}$$

$$o_1 = H_1^{(2)} W_{hq} + b_q =$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$$

7.

8. Assume we have an LSTM for regression, having 2 inputs, a hidden state of dimension 3, and 1 output. For the inputs $X_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$, if the weight matrices and bias vector are generated randomly having values from the set $\{-1, 0, 1\}$ (you can generate them yourself, not all zeros), compute the outputs O_1 and O_2 of the network. Specify the total number of real parameters (weights and biases) for the network.

Formule:

$$I_t = \tilde{\sigma}(x_t W_{xi} + H_{t-1} W_{hi} + b_i)$$

$$F_t = \tilde{\sigma}(x_t W_{xf} + H_{t-1} W_{hf} + b_f)$$

$$O_t = \tilde{\sigma}(x_t W_{xo} + H_{t-1} W_{ho} + b_o)$$

$$\tilde{C}_t = \tanh(x_t W_{xc} + H_{t-1} W_{hc} + b_c)$$

$$C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t$$

$$H_t = O_t \odot \tanh(C_t)$$

Rezolvare: $n = 3$, $d = 2$, $h = 3$, $q = 1$

$X_t \in \mathbb{R}^{n \times d} = \mathbb{R}^{3 \times 2}$ date în problemă

$H_{t-1} \in \mathbb{R}^{n \times h} = \mathbb{R}^{3 \times 3}$ $I_t \in \mathbb{R}^{n \times h} = \mathbb{R}^{3 \times 3}$ $F_t \in \mathbb{R}^{3 \times 3}$ $O_t \in \mathbb{R}^{3 \times 3}$

$W_{xi} \in \mathbb{R}^{d \times h} = \mathbb{R}^{2 \times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ $W_{hi} \in \mathbb{R}^{h \times h} = \mathbb{R}^{3 \times 3} = I_3$

$W_{xf} \in \mathbb{R}^{d \times h} = \mathbb{R}^{2 \times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ $W_{hf} \in \mathbb{R}^{3 \times 3} = I_3$

$W_{xo} \in \mathbb{R}^{d \times h} = \mathbb{R}^{2 \times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ $W_{ho} \in \mathbb{R}^{3 \times 3} = I_3$

$b_i, b_f, b_o \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1 \times 3} = (1 \quad 0 \quad 0)$

param $I, F, O = 3 \cdot 2 \cdot 3 + 3 \cdot 1 \cdot 3 = 18 + 9 = 27$ $+ 3 \cdot 3 \cdot 3$

$\tilde{c}_t \in \mathbb{R}^{m \times h} = \mathbb{R}^{3\times 3}$

$W_{xc} \in \mathbb{R}^{d \times h} = \mathbb{R}^{2\times 3} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$W_{hc} \in \mathbb{R}^{h \times h} = \mathbb{R}^{3\times 3} = J_3$

$b_c \in \mathbb{R}^{1 \times h} = \mathbb{R}^{1\times 3} = (1 \quad 0 \; 0)$

$C_{t-1} \in \mathbb{R}^{m \times h} = \mathbb{R}^{3\times 3} \qquad C_0 = 0^{3\times 3}$

$H_t \in \mathbb{R}^{m \times h} = \mathbb{R}^{3\times 3}$

total param $= 27 + 2\cdot 3 + 3\cdot 3 + 1\cdot 3 = 27 + 18 = 45$

$H_0 \in \mathbb{R}^{m \times h} = \mathbb{R}^{3\times 3} = 0^{3\times 3}$

$$J_1 = ReLu\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + 0^{3\times 3}\cdot J_3 + (1 \; 0 \; 0)\right) =$$

$$= ReLu\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$F_1 = ReLu\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + 0^{3\times 3} J_3 + (1 \; 0 \; 0)\right) =$$

$$= ReLu\left(\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$O_t = \text{ReLu}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} + 0^{3\times 3} J_3 + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \right) =$$

$$= \text{ReLu}\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \right) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\tilde{C}_1 = \tanh\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + 0^{3\times 3} \cdot J_3 + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \right) =$$

$$= \tanh\left(\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \right) = \begin{pmatrix} \tanh(1) & 0 & 0 \\ \tanh(2) & 0 & 0 \\ \tanh(1) & 0 & 0 \end{pmatrix}$$

$$C_1 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \odot \underset{\uparrow}{0^{3\times 3}} + \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \odot \begin{pmatrix} \tanh(1) & 0 & 0 \\ \tanh(2) & 0 & 0 \\ \tanh(1) & 0 & 0 \end{pmatrix} =$$

C_0

$$= \begin{pmatrix} \tanh(1) & 0 & 0 \\ 2\tanh(2) & 0 & 0 \\ \tanh(1) & 0 & 0 \end{pmatrix}$$

$$H_1 = O_1 \odot \tanh(C_1) = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \odot \tanh\left(\begin{pmatrix} \overset{\uparrow 0{,}76}{\tanh(1)} & 0 & 0 \\ 2\overset{\rightarrow 0{,}96}{\tanh(2)} & 0 & 0 \\ \tanh(1) & 0 & 0 \end{pmatrix} \right) =$$

$$= \begin{pmatrix} \tanh(0{,}76) & 0 & 0 \\ 2\tanh(1{,}92) & 0 & 0 \\ \tanh(0{,}76) & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0{,}64 & 0 & 0 \\ 4{,}75 & 0 & 0 \\ 0{,}64 & 0 & 0 \end{pmatrix}$$

$$I_2 = ReLu\left(\left(\begin{pmatrix}1 & 0\\ 1 & -1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{pmatrix} + \begin{pmatrix}0,64 & 0 & 0\\ 4,75 & 0 & 0\\ 0,64 & 0 & 0\end{pmatrix} I_3 + \begin{pmatrix}1 & 0 & 0\end{pmatrix}\right)\right)$$

$$= ReLu\left(\begin{pmatrix}1 & 0 & 0\\ 1 & -1 & 0\\ 0 & 1 & 0\end{pmatrix} + \begin{pmatrix}0,64 & 0 & 0\\ 4,75 & 0 & 0\\ 0,64 & 0 & 0\end{pmatrix} + \begin{pmatrix}1 & 0 & 0\\ 1 & 0 & 0\\ 1 & 0 & 0\end{pmatrix}\right) =$$

$$= \begin{pmatrix}2,64 & 0 & 0\\ 6,75 & 0 & 0\\ 1,64 & 1 & 0\end{pmatrix}$$

$$F_2 = ReLu\left(\left(\begin{pmatrix}1 & 0\\ 1 & -1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix} + \begin{pmatrix}0,64 & 0 & 0\\ 4,75 & 0 & 0\\ 0,64 & 0 & 0\end{pmatrix} I_3 + \begin{pmatrix}1 & 0 & 0\end{pmatrix}\right)\right)$$

$$= ReLu\left(\begin{pmatrix}1 & 0 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix} + \begin{pmatrix}0,64 & 0 & 0\\ 4,75 & 0 & 0\\ 0,64 & 0 & 0\end{pmatrix} + \begin{pmatrix}1 & 0 & 0\end{pmatrix}\right) =$$

$$= \begin{pmatrix}2,64 & 0 & 0\\ 6,75 & 0 & 0\\ 1,64 & 0 & 0\end{pmatrix}$$

$$O_2 = ReLu\left(\left(\begin{pmatrix}1 & 0\\ 1 & -1\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{pmatrix} + \begin{pmatrix}0,64 & 0 & 0\\ 4,75 & 0 & 0\\ 0,64 & 0 & 0\end{pmatrix} I_3 + \begin{pmatrix}1 & 0 & 0\end{pmatrix}\right)\right)$$

$$= ReLu\left(\begin{pmatrix}1 & 0 & 0\\ 1 & -1 & 0\\ 0 & 1 & 0\end{pmatrix} + \begin{pmatrix}0,64 & 0 & 0\\ 4,75 & 0 & 0\\ 0,64 & 0 & 0\end{pmatrix} + \begin{pmatrix}1 & 0 & 0\\ 1 & 0 & 0\\ 1 & 0 & 0\end{pmatrix}\right) =$$

$$= \begin{pmatrix} 2{,}64 & 0 & 0 \\ 6{,}75 & 0 & 0 \\ 1{,}64 & 1 & 0 \end{pmatrix}$$

$$\tilde{C}_2 = \tanh\left(\begin{pmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0{,}64 & 0 & 0 \\ 4{,}75 & 0 & 0 \\ 0{,}64 & 0 & 0 \end{pmatrix} y_3 + \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \right)$$

$$= \tanh\left(\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1{,}64 & 0 & 0 \\ 5{,}75 & 0 & 0 \\ 1{,}64 & 0 & 0 \end{pmatrix} \right) =$$

$$= \begin{pmatrix} 0{,}98 & 0 & 0 \\ 0{,}99 & 0 & 0 \\ 0{,}92 & 0 & 0 \end{pmatrix}$$

$$C_2 = \begin{pmatrix} 2{,}64 & 0 & 0 \\ 6{,}75 & 0 & 0 \\ 1{,}64 & 0 & 0 \end{pmatrix} \odot \underbrace{\begin{pmatrix} \tanh(1) & 0 & 0 \\ 2\tanh(2) & 0 & 0 \\ \tanh(1) & 0 & 0 \end{pmatrix}}_{C_1} +$$

$$\begin{pmatrix} 2{,}64 & 0 & 0 \\ 6{,}75 & 0 & 0 \\ 1{,}64 & 1 & 0 \end{pmatrix} \odot \begin{pmatrix} 0{,}98 & 0 & 0 \\ 0{,}99 & 0 & 0 \\ 0{,}92 & 0 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & 0 & 0 \\ 12{,}96 & 0 & 0 \\ 1{,}24 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 2{,}58 & 0 & 0 \\ 6{,}68 & 0 & 0 \\ 1{,}5 & 0 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 4{,}58 & 0 & 0 \\ 19{,}64 & 0 & 0 \\ 2{,}74 & 0 & 0 \end{pmatrix}$$

$$H_2 = \begin{pmatrix} 2,64 & 0 & 0 \\ 6,75 & 0 & 0 \\ 1,64 & 1 & 0 \end{pmatrix} \circ \tanh\left(\begin{pmatrix} 4,58 & 0 & 0 \\ 19,64 & 0 & 0 \\ 2,74 & 0 & 0 \end{pmatrix}\right) =$$

$$\begin{pmatrix} 2,61 & 0 & 0 \\ 6,75 & 0 & 0 \\ 1,62 & 0 & 0 \end{pmatrix}$$